

Interfacing Cassandra and MongoDB NoSQL with Python

Python has a Cassandra driver in its repository to link and interact with the Cassandra engine. MongoDB and Pymongo can be installed on a system to make the connection between MongoDB and Python. This article will show the reader how to interface these NoSQL databases with Python.



Big Data processing and analytics is one of the key domains addressed on corporate, research as well as academic platforms. There are a number of technologies directly associated with Big Data including fault-tolerant clustering and servers, load balancing, NoSQL databases, NewSQL databases, high performance cloud applications and many others.

In database technology, the NoSQL database systems are widely used in social media applications that have exponentially increasing user interactions. To cope up with applications handling petabytes and yottabytes of data per second, there is a need for high performance and fault-tolerant database systems to execute real-time applications without a single point of delay or failure.

Nowadays, Web based applications use unstructured and heterogeneous data formats, which include video, text, audio, live streaming, wireless signals, images, pixels and many others. As in video, each type of file has a number of formats including WEBM, MPEG, OGG, DRC, RM, MP4, NSV, AVI, MXF, 3GP, WMV, OGG, FLV and others. Similarly, there are lots of formats for images and graphics including EXIF, BPG, GIF, CGM, PNG, ARF, CPT, JPEG, WEBP, PCX, PNM, BMP, U3D, TIFF and many others.

The major issue is the compatibility of the database application with all these file formats, in assorted domains. The adoption, integration and implementation of NoSQL databases comes into play at this juncture. NoSQL

databases can integrate and process any type of file format with higher performance.

There are a number of approaches to classifying NoSQL databases, based on the type of data to be processed and stored.

Column: Accumulo, Druid, Cassandra, Vertica and HBase

Document: Apache CouchDB, Couchbase, Clusterpoint, DocumentDB, MongoDB, MarkLogic, OrientDB, Lotus Notes and RethinkDB

Key-Value: Redis, MemcacheDB, Dynamo, Riak, Aerospike FairCom c-treeACE, FoundationDB, MUMPS and CouchDB

Graph: Allegro, MarkLogic, InfiniteGraph, OrientDB, Neo4J, Virtuoso and Stardog

Apache Cassandra

Apache Cassandra is a prominent and high performance open source database system devised and developed at Facebook, with distributed storage integrated so that a huge amount of data can be processed. It is one of the NoSQL database systems with a higher degree of load balancing, fault tolerance and security without any point of failure.

Apache Cassandra is used in enormous commodity systems and servers because it offers scalability, flexibility and robust support to the number of clusters spread across multiple data centres. It has the inherent properties of synchronous replication with a lower latency rate for all the applications as well as users.